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Inventor(s)	COUGHLIN; Michael J. et al.

VAPOR DRYER WITH INTEGRATED PARTICLE MONITORING

Abstract

An apparatus and method for drying substrates is proved, comprising a containment tank having a rinsing liquid region and a headspace a particle detector, comprising an inlet line extending into the rinsing liquid region of the containment tank; a detection unit configured to receive a flow of liquid from the inlet line therethrough; an outlet line extending from the detection unit and into the containment tank; and a pump connected to one of the inlet line, outlet line or detection unit and configured to cause fluid to be drawn from the containment tank through the inlet line and returned to the containment tank through the outlet line a controller configured to receive information from the detection unit.

Inventors: **COUGHLIN; Michael J. (Livermore, CA), SIN; Garrett Ho Yee (San Jose, CA)**

Applicant: **Applied Materials, Inc.** (Santa Clara, CA)

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Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to the post-polishing cleaning and drying of substrates

Description of the Related Art

[0002] An integrated circuit is typically formed, in part, by the sequential deposition of conductive, semiconductive or insulative layers on a semiconductor substrate, for example a silicon wafer, and by the subsequent processing of the layers.

[0003] One fabrication step involves depositing a filler layer over a non-planar surface, and planarizing the filler layer until projecting portions of the upper or outer surface of the non-planar surface are exposed. For example, a conductive filler layer can be deposited on a patterned insulative layer to fill the trenches or holes in the insulative layer. The filler layer is then polished until the upper or outer surface of the insulative layer, i.e., the area of the underlying layer between the holes or trenches, is exposed. This yields a patterned insulative layer having the conductive filler layer embedded into the trenches or holes, which is exposed at the upper or outer surfaces thereof.

[0004] Chemical mechanical polishing (CMP) is one accepted method of planarization. This planarization method typically requires that the substrate be mounted on a carrier head. The exposed surface (upper or outer surface) of the substrate is placed against a rotating polishing pad. The carrier head provides a controllable load on the substrate to push the surface thereof to be planarized against the polishing pad. A polishing liquid, such as slurry with abrasive particles therein, is supplied to the surface of the polishing pad. For example, cerium oxide can be used as an abrasive particle in the polishing of copper filler layers in CMP. In some cases, the particles are supplied from the pad, which is known as a fixed abrasive pad.

[0005] The slurry of abrasive particles can include cerium oxide particulates and organic additives, and can include other carbon-based residues from the polishing process. These must be removed before other processes can be performed on the substrate, such as the deposition of additional film layers formed over the polished surface of the substrate. To remove these particulates, the substrates can be subjected to a cleaning process that can include the use of harsh oxidizing solvents. For example, a mixture of sulfuric acid and hydrogen peroxide (SPM) can be used in the removal of cerium oxide particulates from the surfaces of a substrate after polishing. SPM cleaning can be performed in a parallel separated mode in which each substrate is placed in a bath in a separate container. The substrates are then subjected to wet scrubbing of the surfaces thereof with a physical contact module such as a brush or roller, followed by immersion in an ultrasonically agitated liquid bath. A final drying step is then performed in a dryer after the cleaning. The desire is for the substrate to exhibit a particle free, dry surface after the drying step.

[0006] The objective of drying the substrate is to ensure a clean, dry substrate comes out of the post polishing cleaning and drying apparatus. However, it is known that particulates can be present on the substrate as it enters the dryer. These particulates may, for example, be slurry particulates or particles of the substrate cleaning apparatus itself, such as particulates generated by the breaking down of the rollers and brushes used to scrub the substrate after the polishing thereof. These particulates can become free of, i.e., detach from, the substrate while it is being dried in the dryer, and accumulate in the liquid used in the drying process. The probability of one or more of the particles in the liquid in the dryer reattaching to a substrate increases as the concentration of the particles in the liquid in the dryer increases. The presence of particles on the substrate after cleaning can result in the manufacture of one or more defective semiconductor die on the substrate.

SUMMARY

[0007] In one aspect, an apparatus for drying substrates, comprises a containment tank having a liquid receiving region and a vapor receiving region a particle detector, comprising an inlet line extending into the liquid receiving region of the containment tank; a detection unit configured to receive a flow of liquid from the inlet line therethrough; an outlet line extending from the detection unit and into the containment tank; and a pump connected to one of the inlet line, outlet line or detection unit and configured to cause fluid to be drawn from the containment tank through the inlet line and returned to the containment tank through the outlet line, and a controller configured to receive information from the detection unit.

[0008] In another aspect, a polishing, cleaning and drying apparatus, comprises at least one physical contact module, a containment tank having a liquid receiving region and a vapor receiving region; a robot arm extendible into the physical contact module and into the containment tank; a particle detector, comprising an inlet line extending into the liquid receiving region of the containment tank; a detection unit configured to receive a flow of liquid from the inlet line therethrough; an outlet line extending from the detection unit and into the containment tank; and a pump connected to one of the inlet line, outlet line or detection unit and configured to cause fluid to be drawn from the containment tank through the inlet line and returned to the containment tank through the outlet line.

[0009] In another aspect, a method of drying a substrate comprises providing a containment tank having a liquid region with a cleaning liquid therein and a vapor receiving region extending over the liquid region; sequentially submersing a plurality of substrates into the cleaning liquid in the liquid region; providing a particle detector, comprising an inlet line extending into the cleaning liquid receiving region of the containment tank; a detection unit configured to receive a flow of liquid from the inlet line therethrough; an outlet line extending from the detection unit and into the containment tank; passing the cleaning liquid through the inlet line and through the detection unit; and determining at least one of the size and shape of any particle detected by the detection unit.

[0010] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other aspects, features and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and may admit to other equally effective embodiments.

[0012] FIG. **1** is a schematic top view of a chemical mechanical polishing system.

[0013] FIG. **2**A-C are schematic diagrams of a vapor dryer with a liquid particle counter attached thereto.

[0014] FIG. **3** is a schematic diagram of the liquid particle counter.

[0015] FIG. **4** is a schematic view of a robotic end effector grasping a substrate.

[0016] Like reference symbols in the various drawings indicate like elements.

[0017] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0018] An embodiment disclosed herein is a vapor dryer configured to dry substrates that have been cleaned in a post-chemical mechanical polishing cleaner. The vapor dryer includes a liquid into which the substrates are submerged and are dried while being lifted out of the liquid during the vapor process. A fluid circuit is configured to pull the liquid from the vapor dryer and determine the presence of particles, the types of particles detected or both. Thus a determination can be made that the particulate of the slurry have not been removed by cleaning, particulates have been generated and become attached to the substrates during the cleaning process or both.

[0019] FIG. **1** illustrates an interior plan view of a chemical mechanical polishing (CMP) system **100**. The system **100** generally includes a factory interface module **102**, input module **104**, a polisher module **106**, and a cleaning module **108**. The four major components are generally disposed and interconnected to provide the CMP system **100**.

[0020] The factory interface **102** includes a support to hold plurality of substrate cassettes **110**, a housing **111** that encloses a chamber, and one or more interface robots **112** within the housing **111**. The factory interface robot **112** generally provides the range of motion required to transfer substrates between the cassettes **110** and the input module **104** and between the cleaning module **108** and the cassettes **110**. Substrates to be processed in the chemical mechanical polishing system are generally transferred from the cassettes **110**, through the housing **111** and into the input module **104** by the interface robot **112**. A transfer robot is provided in the input module **104**. The input module **104** generally facilitates transfer of the substrate between the interface robot **112** and the transfer robot **114**. The transfer robot **114** transfers the substrate between the input module **104** and the polisher module **106**.

[0021] The polisher module **106** generally comprises a transfer station **116**, and one or more polishing stations **118**. The transfer station **116** is disposed within the polishing module **106** and is configured to accept the substrate from the transfer robot **114**. The transfer station **116** transfers the substrate to a carrier head **124** of a polishing station **118** that retains the substrate during polishing.

[0022] The polishing stations **118** includes a rotatable disk-shaped platen having a polishing pad-receiving surface, on which a polishing pad **120** is situated. The platen is operable to rotate about an axis generally perpendicular to the pad-receiving surface thereof. The polishing pad **120** can be a two-layer polishing pad with an outer polishing layer and a softer backing layer. The polishing stations **118** further includes a dispensing arm **122**, to dispense a polishing liquid, e.g., an abrasive slurry, onto the polishing pad **120**. In the abrasive slurry, the abrasive particles can be silicon oxide, but for some polishing processes use cerium oxide abrasive particles. The polishing station **118** can also include a conditioner head **123**, which is actuated into selective contact with the exposed surface of the polishing pad to abrade the surface and maintain the polishing pad **120** at a consistent surface roughness.

[0023] The polishing stations **118** include at least one carrier head **124**. The carrier head **124** is operable to hold a substrate **10** against the polishing pad **110** during a polishing operation. Following a polishing operation performed on a substrate, the carrier head **124** will transfer the substrate back to the transfer station **116**.

[0024] The transfer robot **114** then removes the substrate from the polishing module **106** through an opening connecting the polishing module **106** with the remainder of the CMP system **100**. The transfer robot **114** removes the substrate from the polishing module **106** with the polished surface thereof in a horizontal orientation, and reorients the substrate to position the polished surface in a vertical orientation for placement into the cleaning module **108**.

[0025] The cleaning module **108** generally includes one or more cleaning devices that can operate independently or in concert. For example, the cleaning module **108** can include, from the top to the bottom of the page of in FIG. **1**, an SPM module **128** (described further below), an input module **129**, a physical contact module configured as one or more brush or buffing pad cleaners **131**, **132**, a megasonic cleaner **133**, and a drying module **134**. Other possible cleaning devices include chemical spin cleaners and jet spray cleaners. A transport system, e.g., an overhead conveyor **130** that supports robot arms, can walk or run the substrates from cleaning module device to cleaning module device, as well as insert them into, and remove them from, the cleaning module devices. Briefly, the one or more brush or buffing pad cleaners **131**, **132** are devices in which a substrate can be placed and the surfaces of the substrate are contacted with rotating brushes or spinning buffing pads to remove any remaining particulates. The substrate is then transferred to the megasonic cleaner **133** in which high frequency vibrations (at one or more frequencies) are provided by a transducer to input energy into the cleaning liquid in the megasonic cleaner **133** to produce controlled cavitation in the cleaning liquid to remove particles and by-products of polishing from the substrate. Alternatively, the megasonic cleaner can be positioned before the brush or buffing pad cleaners **131**, **132**. A final rinse can be performed in a rinsing module before transferring the substrate to a drying module **134**.

[0026] Although FIG. **1** illustrates the SPM module **128** as the first cleaning device in the sequence, this is not necessary for actual physical position or order of cleaning operations (although having the cleaning devices in same physical order as the order of operations will be more efficient for throughput). For example, the substrate could be processed by a brush or buffing pad cleaner (e.g., a buff pad), then by the SPM module, then by another brush or buffing pad cleaner (e.g., a rotating brush), and then by a jet spray cleaner.

[0027] As described above, the CMP system **100** transfers the substrates from the polishing module **106** into the cleaning module **108**. Debris from the polishing process, e.g., abrasive particles or organic materials or other by-products of the polishing pad or slurry, can be stuck to the substrates. Some of these materials, e.g., cerium oxide particulates, and organic additives from the polishing module **106**, are difficult to remove with the cleaners **131**, **132**, **133** listed above. Therefore, the substrates are moved to an in-line sulfuric peroxide mixture (SPM) module **128** within the cleaning module **108**. The SPM module shown in FIG. **1** is a module that allows for the in-line SPM cleaning of a number of substrates concurrently. The SPM module **128** includes two containers, a cleaning container **124** and a rinse container **126**. Here, where the SPM module is used, a separate robot dedicated to the loading of substrates thereinto and therefrom may be employed. Upon receiving a signal from a system controller, e.g., when the substrate **10***a* is grasped by a robot arm for the SPM module and inserted into the SPM module, and from the SPM module to a dedicated rinse station for rinsing the SPM chemistry from the substrate.

[0028] After the initial cleaning of a substrate in the SPM module, a robot arm of an overhead conveyor **130**, the robot arm having an end effector **200** thereon, grasps the substrate **10***a* in a vertical orientation and moves it from the rinse station **128** to the input module **129**. From the input module **129**, the substrate is moved in the vertical orientation by the robot arm of the overhead conveyor **130** and then inserted downwardly into to one of the brush or buffing pad cleaners **131**, **132**. Alternatively, a robot arm of the overhead conveyor can move the substrate from the output station **182**, through the input module **129**, and thence into one of the brush or buffing pad cleaners **131**, **132**. After the brushing or buffing of the surface of the substrate, it is lifted by a robot arm of the overhead conveyor **130** from the brush or buffing pad cleaners **131**, **132** and into moved over, and then lowered into, the megasonic cleaner **133** to remove particles, organics and other remaining residues of the polishing process therefrom. The substrate is then lifted out of the megasonic cleaner **133** by a robot arm of the overhead conveyor **130** into then moved over, and lowered into, the drying module **134**. In the drying module **134**, the substrate is submerged, in a vertical orientation, into a liquid, and then pulled (i.e. removed) from the liquid while a drying vapor is directed to the location where the substrate contacts the upper surface of a drying liquid.

[0029] Referring now to FIGS. **2***a* to **2***c*, the drying module **134** is shown schematically to illustrate the basic functionality thereof, and generally includes a containment tank **865** configured to hold a rinsing liquid, for example, deionized water, and a source of drying vapor, for example a mixture of isopropyl alcohol and nitrogen. In FIG. **2***a*, the tank is shown in section from the side thereof. In FIG. **2***b* the tank is shown in plan view, and in FIG. **2***c* the tank is shown connected to a particle detection system. To dry a substrate after the cleaning thereof, the containment tank **865** is filled with a rinsing liquid **866***a* to a level at which the substrate **10***a*, in a vertical orientation, can be fully submerged in the rinsing liquid, while there is sufficient headspace **866***b* above the rinsing liquid region **866***a* for a substrate, in the vertical orientation, to be pulled up and out of the rinsing liquid **866***a* but still be within the confines of the tank **865**. To supply the drying vapor, the containment tank **865** contains a port or a plurality of ports **851***a*, **851***b*, extending through the sidewall thereof at a location above the liquid, through which a plurality of nozzle bars **852** located within the headspace **866***b* are connected to the source of vapor. As seen in the top down view of FIG. **2***b*, the nozzle bar or bars **852** terminate inwardly of the containment tank **865** and include a plurality of nozzles **830** spaced along the length thereof within the headspace. Each nozzle bar **852** extends horizontally over the upper surface of the rinsing liquid **866***a*, and a plurality of nozzles **830**, for example, two, four, six or eight nozzles extend from the outer surface thereof. The nozzles **830** may protrude from their respective nozzle bars **852** at an angle to a sidewall **832** of the containment tank **865** such that they extend downwardly and horizontally. Openings **829** are formed at the distal ends on the nozzles **830** opposite to the nozzle bar **852**. The openings **829** are directed toward a center plane **826** of the containment tank **865** along which the substrate to be dried is lowered into the drying liquid **866** and pulled from the drying liquid **866**. In the two nozzle bars **852***a* and **852***b* shown, a first set of openings **829***a* face the intersection of the center plane **826** of the containment tank **865** with the upper surface of the liquid and a second set of openings **829***b* also face the intersection of the center plane **826** of the containment tank **865** with the upper surface of the liquid, opposite to and facing the first set of openings **829***a* across the center plane **826** of the containment tank **865**. Thus, gas or vapor released through the openings **829***a* and **829***b* is directed toward the intersection of the center plane **826** with the upper surface of the liquid. A substrate **10** passing along the center plane **826** will be impinged by the vapor or gas from the opening **829***a* and **829***b* generally equally on both sides thereof at the location where the opposed sides of the substrate are being lifted out of the drying liquid **866**.

[0030] The distal end of the nozzle bar or bars **852** protrude from the ports **851***a*, **851***b* away from an outer side wall **832** of the containment tank **865**. The nozzle bar or bars **852** are connected at a distal end thereof, external to the containment tank, to a first valve line **853**. The first valve line **853** connects the nozzle bars **852** to a valve **854** at the outlet side thereof. A second valve line **855** is connected to the valve at the inlet side thereof. The second valve **855** line leads to and is in fluid communication with an opening in a vapor source, such as a bubbler **856**. The bubbler **856** is a secondary containment tank located external to the containment tank **865**. The bubbler **856** is configured to hold liquid isopropyl alcohol (IPA) and a secondary gas line **857** extends into the liquid IPA and is connected to a nitrogen source **858**. The nitrogen source **858** provides nitrogen to the bubbler **856** through the gas line **857** at a quantity sufficient to supply IPA and Nitrogen to the containment tank **865** for a Marangoni drying process of the surface of the substrate as it is pulled out of the frying liquid. For example, the IPA may be carried on a small bubble of nitrogen, which is coated with IPA as it passes through the IPA in the bubbler **856**. Thus, IPA and nitrogen are supplied from the bubbler **856** and enter into the containment tank **856** through the nozzle openings **829**.

[0031] Referring to FIG. **2***c*, the containment tank **865** comprises an upper opening for placement of a substrate in the vertical orientation into the containment tank **865**. The opening may be configured as a slit having a length greater that the diameter of the substrates to be dried. The vapor dryer tank cover **861** both covers and surrounds the upper end of the containment tank **865**, but leaves opening **869** through which a substrate **10** may be disposed. The vapor dryer tank cover **861** further comprises a port **805**, the port **805** having at least a first opening and a second opening **810***a*, **810***b* respectively.

[0032] As shown in FIG. **4**, an example of an end effector **200** useful for lowering the substrate **10** into the drying liquid **866***a*, and then lifting it therefrom and into the headspace **866***b* as the IPA dries. Nitrogen vapor is directed at the meniscus of the drying liquid **866** to substrate side surfaces is shown. Here, the end effector **200** is supported from above by the overhead conveyor **130**, which moves the end effector **200**, and lowers it into the ultrasonic cleaner **133** and secures a substrate therein. The overhead conveyor then lifts the substrate **10** out of the ultrasonic tank **133**, moves it over opening **869** on the dryer **134**, and thence lowers the end effector **200** and substrate **10** through the opening **769** and immersed the substrate **10** in the drying liquid **866**. The end effector has a pair of horizontally moveable arms **202**, moveable in the direction of the arrows **208**. Each arm **202** includes an edge grip surface **204** having on or more protrusions configured to contact the circumferential outer wall of the substrate **10**. Each arm **202** is connected to the overhead conveyor **130** via a linkage are **208**. The overhead conveyor **130** controls the horizontal spacing between the linkage arms **206**, and thus between the edge grip surfaces **204** by movement thereof in the direction of arrows **210**. Likewise, the overhead conveyor controls the vertical position of the linkage arms **206**, and thus a substrate **10** secured in the end effector **200**, by simultaneous movement thereof in the direction of arrows **212**.

[0033] The Marangoni process comprises first providing the substrate **10** into the containment tank **865** through the opening **869** by the end effector **200** clamping the outer perimeter of the substrate **10**. Next, the substrate **10** in a vertical orientation is dipped into the drying liquid **866***a*, for example, deionized water, so that the substrate **10** is fully submerged in the drying liquid **866***a*. The substrate **10** is then lifted from the drying liquid **866***a* and into the headspace **866***b*, as the substrate **10** is lifted out of the drying liquid **866***b;* IPA and Nitrogen are sprayed from the nozzles at the location of the intersection of the substrate surface with the drying liquid **866**, i.e., the meniscus. When the substrate is pulled up through the stream of IPA and Nitrogen sprayed from the nozzles, the IPA displaces the drying liquid **866** on the surface of the substrate and the surface of the substrate **10** is dried.

[0034] Referring now to FIG. **3**, a Liquid Particle Counter (LPC) module **850** is provided herein to enable the monitoring of the condition of the drying liquid **866** in the containment tank **856** with respect to the concentration and size or type of particulates in the drying liquid **866***a*. The LPC module **850** includes an Ethernet line **922** connected to and from a controller having a processing function and a memory coupled to the processing function **932**, a first power supply line **921***a* and a second power supply line **921***b* connected to a power supply **931**, a sample inlet line **801** connected from the containment tank **865** and a sample outlet line **802** connected within the containment tank **865**. The sample inlet line **801** may extend through a wall of the containment tank **865** or may be flush with the wall of the containment tank **865**. A signal line **942** connects the controller and the power supply **931**. The sample outlet line **802** is provided through the second opening **810***b* in the port **805** on the cover **804** of the containment tank **865** and extends within the containment tank **865** to a point below the upper surface of the drying liquid **866** in the containment tank **865**. The sample outlet line **802** may also extend to a point above to fluid line in the containment tank **865**. A pump **927** is connected to the sample inlet line **801** and when powered sucks drying liquid **866***a* from the containment tank **865**, through a detection section of the LPC, and then pushes the liquid through a flow meter **926** and thence back to the containment tank **865** through the sample outlet line **802**. As shown in FIG. **2***c*, the sample inlet line **801** extends through a first opening **810***a* in the a port **805** in the cover **804** of the tank **865**, and the sample outlet line **802** extends through a second opening **810***b* in the port **805** on the cover **804** of the containment tank **865**. The Ethernet line **903** connects to the controller **932** at the distal end thereof enters the LPC unit and connects to a photodiode **903** at the proximal end thereof. A controller line **940** connects the controller **932** to an operation manager **941** operable to control the fluid circuit of the sample inlet line **801**, detection section and sample outlet line **802** based on information sent to the controller **932**. The first power supply line **921***a* connected from the power supply **931** at the distal end thereof, enters the LPC unit and connects to the photodiode **903** at the proximal end thereof. The second power supply line **921***b* connected from the power supply **931** at the distal end thereof, enters the LPC unit and connects to a laser **901** at the proximal end thereof.

[0035] The first power supply line **921***a* connected from the power supply **931** at the distal end thereof also connects to a third power supply line **921***c* at a point on the first power supply line

921*a* between the photodiode **903** and the power supply **931**. The third power supply line **921***c* connects from the first power supply line **921***a* to a speed (flow rate) control unit **925**. A fourth power supply line **921***d* connects the speed control unit **925** to the pump **927** and the flow meter **926**, thus the pump **927**, flow meter **926**, and the speed control unit **925** are connected to the power supply **931**. The pump **927** and the flow meter **926** are connected at a portion of the sample outlet line **802** at a point between the Containment tank **865** and the LPC unit.

[0036] The LPC module **850** includes the LPC unit **900** comprising a flow tube **902**, the laser **901**, and a photodiode **903**. The LPC unit comprises an outer unit container **908** having a first opening **910**, a second opening **911** opposed to the first opening **910**, a third opening **913**, a fourth opening **914**, and a fifth opening **915** therein. The flow tube **902** includes a first end **906** and a second end **907**, and the flow tube **902** spans between and is sealingly connected through the first opening **910** to the sample inlet line **801** and through the second opening **911** to the sample outlet line **802**. The first opening **910** is in fluid communication with the first end **906** of the flow tube **902** and the second end **907** of the flow tube **902** is in fluid communication with the second opening **911**. The distal end of the sample end line **801** is also in fluid communication with the first opening **910**, and the sample outlet line **802** is in fluid communication with the send opening **911**. The pump **927** including the flow meter **926** controls the flow rate of fluid through the sample inlet line **801**, the flow tube **902**, and the sample outlet line **092**. The flow tube **902** is composed of a transparent material that is not reactive with the liquid flowing therethrough, such as glass or a polyacrylate, thereby allowing light to enter and exit the flow tube during flow of fluid therein.

[0037] The laser **901** in the LPC unit uses a LASER diode light source and LASER beam shaping optics to illuminate a cross section of the liquid flow path within the flow tube **902**. As the drying liquid **866***a* flows through the flow tube **902**, it passes through a LASER beam emitted by the laser **901**. When particles are in the flow stream of the drying liquid **866** passing through the laser beam and alighted by the laser, the signature of the intensity of the light passing through the tube and exiting the other side of the flow tube **902** changes. For example, the light can be absorbed, scattered, or partially absorbed and partially scattered by the particle. The light passing through the flow tube **902** is collected by an optical system **904** on the opposite side of the flow tube **902** to the entry location of the laser beam of the laser **901** into the flow tube **902** and imaged onto a photodiode. The photodiode converts the light into an electrical current, which is converted to voltage and amplified. The signal sent by the photodiode exits the photodiode through the Ethernet line **922**, which is linked to the controller **932**.

[0038] The result is a voltage pulse each time a particle crosses the LASER beam. The width of the pulse is proportional to the time it takes the particle to cross the LASER beam and the pulse's amplitude is proportional to the size of the particle. The voltage pulses created by the particles are processed by additional electronics to quantify the pulses by the size of each particle. The quantities of the various size particles are processed and stored in the sensor's buffers or transferred via the Ethernet line to the controller **932**.

[0039] The controller **932** can contain software that generates a graphical display of information to a user via a user interface regarding particle size and count. The controller may also contain software that will alert a user to issues with the cleaning process based on the particle size and/or the particle count reaching a certain threshold or range of thresholds. For example, if slurry particles are detected in the vapor dryer tank **865**, then a notification is sent to a service engineering team that there is an issue with cleaning in the initial clean, the brush boxes or the megasonic clean system (if applicable).

[0040] The Software in the controller can be configured to provide a number of different items of information useful to the user of the system. As discussed, it can be configured to raise an alarm when the quantity of particles detected in a predefined quantity of drying liquid **866***a* passing through the LPC meets or exceeds a threshold number of particles. In this scenario, the quantity of particles in a predefined quantity of fluid is indicative of the particle density in the drying liquid

866*a*. The system can be configured to raise an alarm when the quantity of particles of a certain size or range of sizes detected in the predefined quantity of drying fluid **866***a* meets or exceeds a threshold number of particles. Likewise, it can be configured to raise different alarms for different sizes or quantities of particles. For example, the LPC can be calibrated using particles of a size of the abrasive particles in the slurry, and based on a threshold number of such particles being detected in a predetermined quantity of drying liquid **866** flowing through the LPC, raise an alarm that the cleaning elements of the system prior to the drying thereof in the dryer **134** is not adequately removing the slurry from the substrates. Additionally, particles of the cleaning components such as the brushes or rollers in the brush box may have different sizes and different outlines of the shadow cast by them as they pass over the photodetector. Thus, when particles of the configuration that would be shed by a brush or roller when the brush or roller is deteriorating can be detected, and an alarm specific to the brush box needing servicing can be provided.

[0041] The operation manager **941** receives input from the controller **932** about how to process drying liquid **866***a* through the LPC module **850** circuit. The controller may for example, signal to the system to begin pumping the drying liquid **866***a* through the LPC module by user input or system input signaling that the start of sampling of the drying liquid **866***a* is needed. A signal is sent to the controller **932** through the signal line **942**, and the controller turns on the power supply **932**. Power is supplied to the speed control, the pump **927** and flow meter **926**, as well as the photodiode **903** in the LPC unit **900**. The pump **927** pulls drying liquid **866***a* from the Containment tank **865**, through the LPC unit **900**, and outputs the drying liquid **866***a* back into the Containment tank **865**. The controller will send a signal to the speed control **925** to cause the fluid to be pumped more or less quickly, i.e., to modify the flow rate of the drying liquid **866***a* through the LPC, or to start and stop flow of the drying liquid **866***a* through the LPC to create in pulses of drying liquid **866***a* through the LPC unit **900**. The controller sends a signal to the controller **932** either to engage in particle detection in the drying liquid **866***a* continuously as substrate processing proceeds or to engage in particle detection in the drying liquid **866***a* in the Containment tank **865** intermittently. The particle detection in the drying liquid **866***a* may be performed intermittently by receiving a signal from the controller that a substrate is currently being processed, and only perform particle detection in the drying liquid **866***a* before, during, or after a substrate is being processed or a combination thereof. The particle detection in the drying liquid **866***a* may be performed intermittently by receiving a signal from the controller to perform particle detection in the drying liquid **866***a* in time spaced pulses regardless of whether a substrate is being processed. Likewise, particle detection in the drying liquid **866***a* may be performed continuously.

[0042] The controller **932** receives information from the photodiode about the size of the particles processed in the LPC module **900**. The controller **932** receives information from the photodiode about the shape of the particles processed in the LPC module **900**. The controller **932** uses software to determine if the number of the particles being processed through the LPC module per unit of time reach an alarm threshold. An alarm will be triggered if the particles per unit of time reach a certain threshold. The alarm can be visual, audio, or an electronic message. The information provided to the controller about the size and shape of the particles detected can be analyzed by the software and provide a user with information about possible issues. For example, larger particles may indicate a break down in the brush box, or smaller particles may indicate issues with a slurry formula or cleaning.

[0043] A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. An apparatus for drying substrates, comprising: a containment tank having a rinsing liquid region and a headspace; and a particle detector comprising: an inlet line; a liquid particle counter configured to receive a flow of liquid from the containment tank via the inlet line and to output information about particles in the liquid; an outlet line; and a pump configured to cause the liquid to be drawn from the containment tank into the liquid particle counter through the inlet line and returned to the containment tank through the outlet line.
2. The apparatus of claim 1, further comprising a controller operable to control the flow of liquid through the particle detector.
3. The apparatus of claim 1, further comprising a speed control unit operable to control a rate of the flow of liquid.
4. The apparatus of claim 1, wherein the outlet line has a fluid outlet connected to the rinsing liquid region of the containment tank.
5. The apparatus of claim 1, further comprising a controller configured to discriminate particles based on size from the information provided from the liquid particle counter.
6. The apparatus of claim 1, further comprising a controller configured to output an alarm based on the information obtained from the liquid particle counter.
7. The apparatus of claim 1, further comprising a controller configured to discriminate particles based on shape from the information provided from the liquid particle counter.
8. A polishing, cleaning and drying apparatus, comprising: a polishing module to polish a substrate; a physical contact module to clean the substrate after the polishing; a vapor dryer to dry the substrate after the substrate is cleaned by submerging the substrate in rinsing liquid of a rinsing liquid region and drying the substrate as the substrate is removed from the rinsing liquid; and a liquid particle counter to receive a flow of the rinsing liquid from the vapor dryer and to output information about particles in the rinsing liquid.
9. The apparatus of claim 8, further comprising a controller configured to control the flow of the rinsing fluid through the liquid particle counter.
10. The apparatus of claim 8, further comprising a speed control unit configured to control a rate of the flow of the rinsing fluid through the liquid particle counter.
11. The apparatus of claim 8, further comprising an outlet line having a fluid outlet that is disposed in the rinsing liquid region.
12. The apparatus of claim 8, further comprising a controller configured to discriminate particles based on size from the information provided from the liquid particle counter.
13. The apparatus of claim 8, further comprising a controller configured to output an alarm based on the information obtained from the liquid particle counter.
14. The apparatus of claim 8, further comprising a controller configured to discriminate particles based on shape the information provided from the liquid particle counter.
15. A method of substrate processing, the method comprising: polishing a substrate; after the polishing, cleaning the substrate; and after cleaning the substrate, submerging the substrate into a rinsing liquid and drying the substrate as the substrate is removed from the rinsing liquid; and passing the rinsing liquid through a liquid particle counter.
16. The method of claim 15, further comprising returning the rinsing liquid to a rinsing liquid region through an outlet line.
17. The method of claim 15, further comprising continuously passing the rinsing liquid through an inlet line and through the liquid particle counter.
18. The method of claim 15, further comprising intermittently passing the rinsing liquid through an inlet line and through the liquid particle counter.
19. The method of claim 15, further comprising setting off an alarm based on information obtained from the liquid particle counter.

20. The method of claim 19, wherein the alarm is one of a visual, audio, or an electronic message alarm.